

ZkCertify: Bachelor’s Degree Certificate Generation and Verification Using Zk-proofs

Abstract—Traditional university systems manage student grades, departmental clearances, and graduation certificates across separate spreadsheets and paper records. This manual process causes verification delays and forces graduates to expose their entire transcript when employers only need to verify basic facts, such as degree completion or a minimum CGPA. We introduce ZkCertify, a unified blockchain platform that handles the entire academic lifecycle from semester grade calculation to instant public verification. To keep transaction costs under \$0.02, ZkCertify records only a 32-byte cryptographic hash on the Polygon Amoy blockchain, while storing certificate PDFs on IPFS and active records in a standard database. To protect student privacy, graduates can use browser-based zero-knowledge proofs (zk-SNARKs) to prove specific academic qualifications without revealing individual course grades. In addition, identity verification runs locally in the browser using ArcFace neural face matching, so personal photos are never uploaded to a remote server. The platform also automates semester GPA calculations, tracks five institutional clearances, and supports an auditable replacement process for official grade corrections.

Index Terms—Academic credentials, blockchain, Polygon, IPFS, smart contracts, Keccak-256, zk-SNARK, zero-knowledge proofs, result calculation, semester grade sheets, clearance management, credit transfer, emergency leave, face recognition.

I. INTRODUCTION

Academic credentials play a key role in employment, postgraduate admissions, professional licensing, and global mobility. The authenticity of a degree certificate depends on the entire academic process: recording assessment marks, computing cumulative GPAs, completing departmental clearances, and securing authorized approvals. In most universities, these tasks are handled by separate administrative offices using a mix of local databases, spreadsheets, paper forms, and physical approval queues [1], [2].

This fragmented workflow creates four practical problems:

- 1) **Tampering Risks:** Standard digital PDF certificates can be modified without visible visual flaws, forcing employers to rely on manual confirmations from the issuing university.
- 2) **Administrative Overhead:** Calculating semester GPAs, tracking backlog courses, and collecting physical clearance stamps across departments requires repetitive manual effort.
- 3) **Disconnected Records:** Approved emergency leaves and transferred course credits are often stored in isolated files, making final degree audits difficult to reconstruct.
- 4) **Privacy Over-Disclosure:** Standard verification requires graduates to share their entire transcript, even when an employer only needs to verify degree completion or a minimum CGPA [3].

Public blockchains provide an immutable integrity record for credentials. However, storing complete certificates or student records directly on-chain is expensive and exposes personal data. In addition, universities occasionally need to issue legitimate grade corrections after an official appeal; the system must support these revisions without silently overwriting previous records [2], [4], [5].

We built ZkCertify around these operational requirements. The framework separates immutable integrity records from document storage and live operational data, while connecting certificate issuance directly to the university workflow that creates the record. It also supports selective verification using browser-based zero-knowledge proofs and runs facial matching locally to protect student privacy. Our main contributions are:

- A three-tier hybrid storage model that anchors 32-byte Keccak-256 commitments on Polygon Amoy, pins credential files to IPFS, and maintains operational data in MySQL via Prisma.
- An automated academic result engine that computes credit-weighted SGPA and cumulative CGPA, flags academic probation or backlogs, and produces semester grade sheets.
- A digital clearance workflow spanning accounts, library, residential halls, laboratories, and examination divisions, with support for emergency leave and transfer credit records.
- Client-side Groth16 [6] zk-SNARK circuits for proving selected credential claims (CGPA thresholds, degree completion, 160.0 credit hours, alumni status) without disclosing raw course grades.
- A browser-based ArcFace [7] biometric verification module that matches 512-dimensional facial embeddings locally on the user’s device.
- A role-based governance model covering students, minters, signatories, administrators, and verifiers, complete with an on-chain burn-and-reissue procedure for auditable corrections.

II. RELATED WORK

Researchers have evaluated blockchain-based credential systems on both permissioned and public networks. Indhuja *et al.* [8] automated certificate issuance on Hyperledger Fabric, showing that smart contracts reduce manual checks. Pawar *et al.* [4] paired blockchain hashes with IPFS storage to avoid on-chain file storage bottlenecks.

TABLE I
COMPARISON WITH RELATED ACADEMIC CREDENTIAL SYSTEMS.

Approach	Ledger	Storage	Privacy	Verification	Scope
Indhuja et al. [8]	Hyperledger Fabric	On-chain	Not Reported (NR)	Smart-contract validation	Certificate issuance and verification
Pawar et al. [4]	Blockchain	IPFS	NR	Certificate verification	E-certificate generation and verification
Hariprasath and Kumar [3]	Dual blockchain	NR	ZK proofs and threshold signatures	Privacy-preserving verification	Selective certificate verification
Chowdhary et al. [9]	Blockchain	NR	Biometric descriptors	Student identity binding	Student identity and certificate verification
Rani et al. [1]	Hyperledger Fabric	CouchDB	NR	Digital signatures	Permissioned certificate notarization
Farabi et al. [10]	Ethereum	IPFS	NR	QR and smart contract	Role-based credential issuance
ZkCertify	Polygon Amoy	IPFS + MySQL	Groth16 ZK + local face matching	Hash, QR, document, and claim checks	Academic lifecycle and selective verification

Ethereum-based frameworks have also been investigated. Sree [5] proposed decentralized certificate issuance using Ethereum smart contracts, while Jaafar *et al.* [11] used Ethereum alongside IPFS to verify educational credentials. Noorhizama *et al.* [12] introduced QR-based verification for Ph.D. certificates, simplifying credential lookup. Kadam *et al.* [13] developed ValidCertify for domain-specific authentication, and Rani *et al.* [1] built EduCert-Chain around permissioned notarization. Deepika *et al.* [14] analyzed blockchain verification for student record integrity. Sultana *et al.* [15] formalized an IPFS-blockchain smart contract architecture designed to curb academic credential fraud. Farabi *et al.* [10] developed ShikkhaChain, addressing certificate fraud in Bangladesh through Ethereum, IPFS, and role-based governance.

Privacy-preserving verification remains an active research area. Hariprasath and Kumar [3] used threshold signatures and zero-knowledge proofs on dual blockchains so certificate properties can be checked without revealing private fields. Chowdhary *et al.* [9] applied biometric descriptors to bind credentials directly to student identity.

Most existing studies treat certificate issuance and verification as isolated tasks, giving little attention to the operational steps that precede issuance, such as repeated result computation, departmental clearances, leave tracking, and credit transfer. Rustemi *et al.* [2] highlighted this limitation in their systematic review. ZkCertify addresses this gap by integrating student assessment, multi-office clearances, and selective credential proofs into a unified platform (Table I).

III. SYSTEM ARCHITECTURE AND DESIGN

ZkCertify uses a modular architecture that separates ledger state, document storage, institutional operations, and client-side privacy computations (Fig. 1).

A. Hybrid Storage and Data Model

Data is partitioned across three storage tiers according to access needs and gas efficiency (Table II):

- 1) **Polygon Amoy Blockchain:** Acts as the public integrity anchor. It records only the 32-byte credential hash, authorized issuer address, issuance timestamp, credential type, and revocation status.
- 2) **IPFS File Storage (Pinata):** Stores full certificate PDFs, semester grade sheets, canonical metadata JSON, and reference photographs.
- 3) **Operational Database (Prisma/MySQL):** Handles active academic data, continuous assessments, clearance records, emergency leave entries, transfer credit mappings, and administrative audit logs.

TABLE II
STORAGE LAYER PARTITIONING.

Layer	Purpose	Representative Data
Polygon Amoy	Integrity anchor	Hash, issuer, time, status
IPFS / Pinata	Document storage	PDF, grade sheets, photo, JSON
Prisma / MySQL	Operations	Courses, clearance, leave, transfer
Client browser	Local computation	zk-SNARK proofs, face matching

B. Credential Issuance and Verification Logic

The issuance pipeline validates institutional records, serializes data deterministically, calculates its hash digest, and submits the commitment on-chain. Verification reconstructs the digest and checks it against the smart contract registry. Algorithm 1 and Algorithm 2 summarize these protocols.

C. Positioning and Novelty

Existing systems primarily focus on post-graduation verification. ZkCertify extends this scope by connecting continuous result calculation, departmental clearances, signatory governance, credential issuance, revision, and privacy proofs into one integrated lifecycle. The framework cleanly separates on-chain integrity commitments from off-chain document storage and client-side zero-knowledge proofs.

D. Deterministic Cryptographic Commitment and Notation

The system parameters and mathematical notation are summarized in Table III. A credential commitment is generated

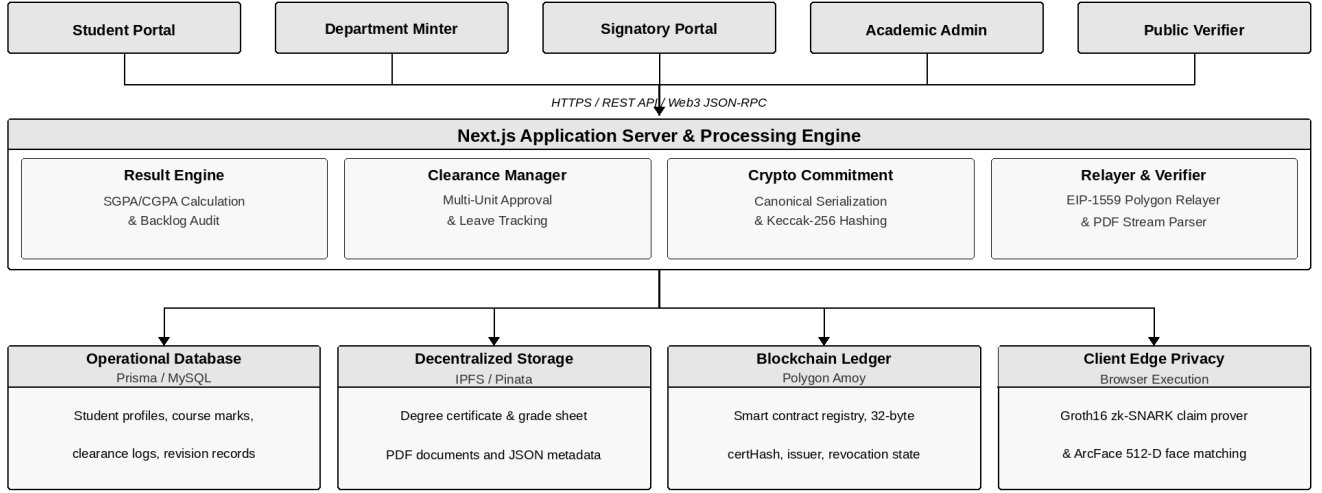


Fig. 1. System architecture of ZkCertify.

Algorithm 1 Credential issuance

Require: Approved academic record R , authorized issuer I

Ensure: On-chain credential commitment H

- 1: Validate R and required institutional approvals
- 2: **if** validation fails **then**
- 3: **return** REJECT
- 4: **end if**
- 5: $P \leftarrow \text{CanonicalSerialize}(R)$
- 6: $H \leftarrow \text{Keccak256}(P)$
- 7: Submit (H, I) to `issueCertificate`
- 8: **if** transaction is confirmed **then**
- 9: Store certificate artifact and metadata in IPFS
- 10: **return** H
- 11: **else**
- 12: **return** RETRY
- 13: **end if**

Algorithm 2 Credential verification

Require: Credential artifact D or credential hash H

Ensure: Verification result

- 1: **if** H is not supplied **then**
- 2: $R \leftarrow \text{ExtractCanonicalRecord}(D)$
- 3: $H \leftarrow \text{Keccak256}(\text{CanonicalSerialize}(R))$
- 4: **end if**
- 5: Query registry with H
- 6: **if** record exists and is not revoked **then**
- 7: **return** VALID
- 8: **else**
- 9: **return** INVALID
- 10: **end if**

by computing the Keccak-256 hash of the canonically concatenated academic record:

$$\begin{aligned} \text{payload} &= \text{name} \parallel \text{roll} \parallel \text{reg} \parallel \text{dept} \\ &\quad \parallel \text{degree} \parallel \text{cgpa} \parallel \text{year} \parallel \text{type}, \\ \text{certHash} &= \text{Keccak256}(\text{payload}). \end{aligned} \quad (1)$$

For semester grade sheets, the payload additionally includes the semester number, registered credits, and earned SGPA. Because the serialization is deterministic, any altered field changes the resulting hash and fails verification immediately.

TABLE III
SYSTEM NOTATION.

Symbol	Meaning	Value / Domain
H	Credential hash	Keccak-256, 32 bytes
$\mathcal{R}$	Credential registry state	On-chain (Polygon Amoy)
C_i	Course credit value	Curriculum defined
GP_i	Course grade point	Curriculum defined
δ_{th}	Face-match threshold	Configured experimentally
N	Supported degree credits	160.0 credit hours
T_{zk}	ZK verification time	< 1.0 s in prototype test
G_{inv}	Credential invocation gas	45,000 to 65,000 gas
G_{dep}	Contract deployment gas	Not measured
TPS	End-to-end throughput	Not measured
L_{conf}	Confirmation latency	Not measured

E. Credential State Transition

Let S denote the credential state, where $S \in \{\text{unissued}, \text{active}, \text{revoked}\}$. Issuance transitions an approved record from unissued to active. Revocation transitions an active record to revoked. An authorized correction is minted as a new commitment rather than overwriting historical blocks:

$$S_{t+1} = \begin{cases} \text{active}, & S_t = \text{unissued}, \text{ issue succeeds,} \\ \text{revoked}, & S_t = \text{active}, \text{ revoke succeeds,} \\ S_t, & \text{otherwise.} \end{cases} \quad (2)$$

This state model preserves full traceability for administrative corrections.

F. Smart Contract Architecture

The core registry contract, `CertificateRegistry`, keeps on-chain storage minimal:

$$\mathcal{R} = \{\text{certHash} \in \mathbb{B}^{32}, \text{issuer} \in \mathbb{A}, \text{issuedAt} \in \mathbb{N}, \text{isRevoked} \in \mathbb{B}\}. \quad (3)$$

Here, $\mathbb{B}^{32}$ is the 32-byte hash space, $\mathbb{A}$ authorized addresses, and $\mathbb{B}$ Boolean values. A revision counter is tracked off-chain in the database to link subsequent updates. Table IV lists the smart contract functions.

TABLE IV
CONTRACT FUNCTIONS.

Function	Access	Purpose
<code>issueCertificate</code>	Issuer	Records credential hash
<code>revokeCertificate</code>	Issuer	Marks credential revoked
<code>verifyCertificate</code>	Public	Returns credential status
<code>authorizeIssuer</code>	Owner	Grants issuer permission

The end-to-end credential lifecycle is shown in Fig. 2.

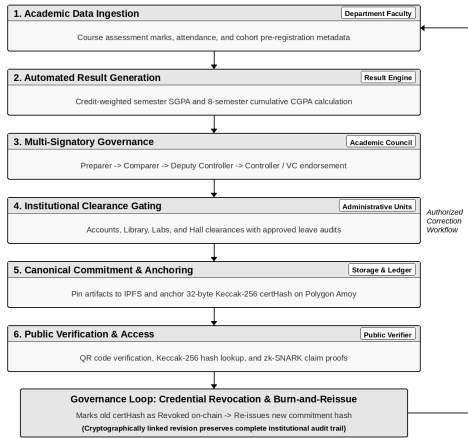


Fig. 2. Certificate lifecycle workflow.

IV. METHODOLOGY AND MULTI-PORTAL WORKFLOW

ZkCertify separates administrative responsibilities across five role-specific portals to prevent account over-privileging:

A. Student Portal

Students sign in using their institutional email with six-digit time-based OTP verification. The student portal provides access to the following functions:

- **Grade sheet retrieval:** View and download QR-enabled PDF grade sheets for completed semesters.
- **Clearance tracking:** Monitor live approval status from departmental laboratories, the central library, residential halls, accounts, and the examination controller.

- **Administrative requests:** Submit records for approved emergency academic leave and requests for credit transfer recognition.
- **Zero-knowledge proof generation:** Generate browser-side proofs for credential claims without publishing raw transcripts.

B. Department Minter Terminal and Result Generation Engine

Department faculty enter student assessment marks using structured CSV or Excel spreadsheets. The engine calculates the credit-weighted SGPA for each semester:

$$\text{SGPA} = \frac{\sum_i C_i \times GP_i}{\sum_i C_i}, \quad (4)$$

where C_i is course credit and GP_i is grade point. The engine computes cumulative CGPAs across eight semesters, checks for probation or backlogs, and produces official grade sheets. Approved emergency leaves are recorded to prevent interruptions from appearing as academic failures. The transfer module maps approved external courses directly into the graduation audit.

C. Signatory Portal

Degree certificate issuance follows a configurable multi-stage review pipeline: Preparer $\rightarrow$ Comparer $\rightarrow$ Deputy Controller $\rightarrow$ Controller/Vice-Chancellor. Each signatory must digitally sign the candidate batch before it advances to blockchain anchoring.

D. Academic Admin Controller Portal

The administration portal coordinates institution-wide batch issuance. Degree minting is gated until all five departmental clearances are verified. Once cleared, an authorized administrator submits the candidate batch to Polygon Amoy using the institutional Web3 wallet. The portal also manages minter permissions and audit logs.

E. Public Verification Portal

External verifiers can inspect certificate without an account (Fig. 3). The system supports four verification channels: direct 32-byte hash lookup, scanning the printed QR code, uploading the certificate PDF for OCR extraction, or validating a zk-SNARK claim proof.

F. Zero-Knowledge Proof and Biometric Modules

1) **zk-SNARK Selective Verification:** Using browser-based Groth16 [6] circuits, graduates can generate mathematical proofs for four claims without revealing individual course grades:

- 1) **CGPA threshold:** Proves $\text{CGPA} \geq \tau$, where $\tau \in \{3.00, 3.25, 3.50, 3.75, 3.80\}$.
- 2) **Degree completion:** Proves graduation with a B.Sc. in Engineering from Sylhet Engineering College.
- 3) **Credit requirement:** Proves completion of the required 160.0 credit hours.
- 4) **Institutional affiliation:** Proves registered institutional affiliation.

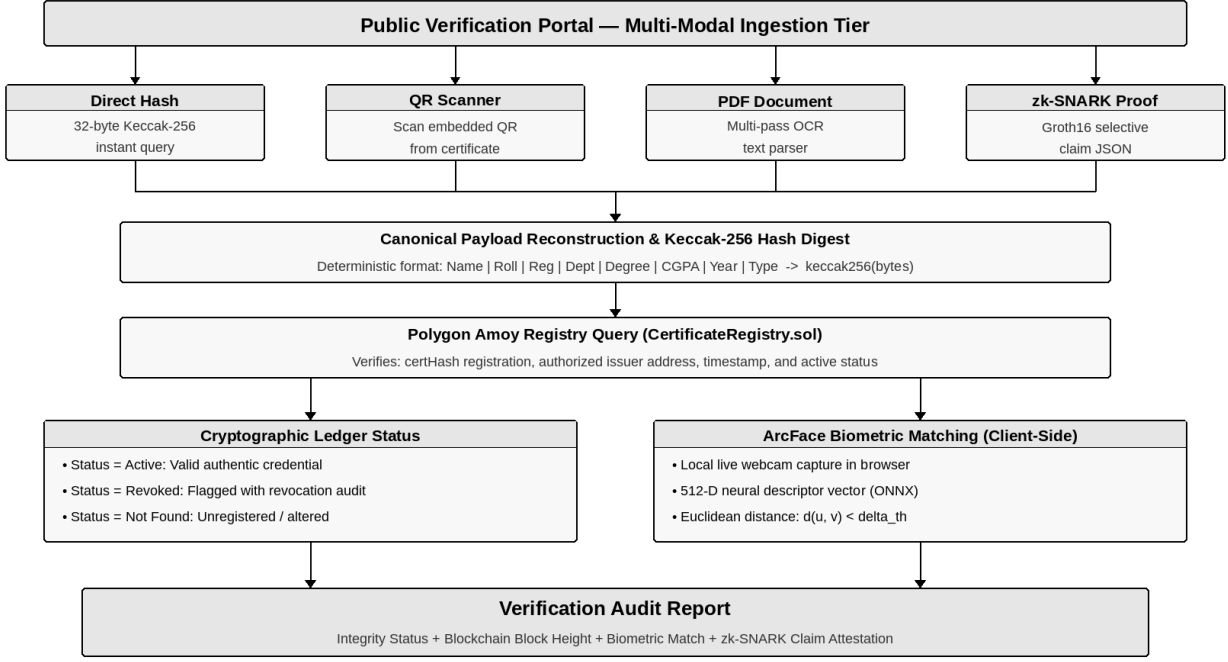


Fig. 3. Public verification framework.

In prototype testing, client-side proof generation and verification completed in under one second.

2) *Client-Side Biometric Verification*: To prevent credential sharing, an ArcFace deep neural network [7] extracts a 512-dimensional facial embedding ($\mathbf{u}$) from the student's live camera and compares it against the reference embedding ($\mathbf{v}$):

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{\sum_{i=1}^{512} (u_i - v_i)^2} < \delta_{th}. \quad (5)$$

Because matching runs entirely in the browser using WebAssembly and WebGL, raw facial images and embeddings are never sent across the network.

V. IMPLEMENTATION AND EVALUATION

A. Implementation Stack

The prototype frontend is built with Next.js 15 (App Router), TypeScript, and Tailwind CSS. Blockchain connectivity uses Wagmi v2 and Viem. Smart contracts are written in Solidity 0.8.24 and deployed to Polygon Amoy using Hardhat. Files are pinned to IPFS via Pinata, and operational records are managed in MySQL via Prisma ORM.

B. Gas and Runtime Evaluation

By anchoring only 32-byte commitments, the system keeps on-chain costs minimal. Individual credential minting consumes between 45,000 and 65,000 gas units, costing approximately \$0.01 to \$0.02 on the testnet. Table V summarizes the benchmark results.

The current evaluation demonstrates functional feasibility. Full production deployment will require further testing under

TABLE V
PROTOTYPE EVALUATION AND PERFORMANCE.

Criterion	Observed Result
Integrity	Modified records produce hash mismatch
Authenticity	Unauthorized issuers rejected
Result engine	500+ students in 1.4 s
Clearance	Minting blocked until approved
Credit transfer	External credits mapped to audit
Revocation	Revoked status retained on-chain
ZK verification	Proof verified in < 1 s (client)
Biometrics	Matching performed in browser
Invocation gas	45,000 to 65,000 gas per credential
On-chain storage	32-byte hash per credential
Document storage	IPFS, off-chain
Deployment gas	Not measured
Throughput (TPS)	Not measured
Confirmation latency	Not measured

high concurrency and comprehensive biometric false acceptance and rejection rate evaluations across diverse demographic datasets.

C. Governance Lifecycle: Revocation and Reissuance

When an official academic appeal requires a grade correction, the administrator invokes `revokeCertificate` to flag the old hash as revoked on Polygon Amoy. A new hash is computed from the revised record and minted on-chain. The relational database links both entries with an incremented revision counter, preserving an honest and auditable correction history.

VI. DISCUSSION

ZkCertify separates integrity anchoring, document storage, and operational data. Anchoring only 32-byte hashes eliminates blockchain bloat while guaranteeing tamper-proof verification.

Selective verification changes credential checking from full transcript disclosure to targeted claim verification. The zk-SNARK module enables graduates to prove academic milestones without exposing private course marks. Similarly, running ArcFace facial matching locally in the browser protects biometric privacy while confirming holder identity.

Because empirical tests were conducted on the Polygon Amoy testnet, production deployments must account for mainnet base fee dynamics, key rotation, wallet recovery procedures, and IPFS pin maintenance.

VII. CONCLUSION AND FUTURE WORK

ZkCertify presents an integrated academic credential management platform that connects semester grade calculation, departmental clearances, IPFS storage, on-chain integrity anchoring, zero-knowledge proofs, and local biometric matching. The architecture ensures certificate authenticity while protecting student privacy. Future work will investigate mobile wallet integration, W3C Verifiable Credentials [16], and hardware-assisted liveness detection.

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